MILITARY SPECIFICATION

FENDERS, MARINE, FOAM FILLED, NETLESS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high energy absorption elastomeric marine fender system to be used for protection of ships, harbor craft, wharves, and piers from damage between the interface of vessel to vessel or vessel to pier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILIARY

MIL-C-3774 - Crate, Wood, Open, 12,000 and 16,000 Pound Capacity.
MIL-I-45208 - Inspection System Requirements.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2040

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(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Steel, Structural.
A123 - Zinc (Hot Galvanized) Coatings on Products Fabricated From Rolled, Pressed, and Forged Shapes, Bars, and Strips.
A153 - Zinc Coating (Hot Dip) On Iron and Steel Hardware.
D395 - Rubber Property in Compression, Test for.
D412 - Rubber Properties in Tension, Test Methods for.
D470 - Thermosetting Insulated and Jacketed Wire and Cable.
D624 - Rubber Property Tear Resistance, Test for.
D750 - Rubber Deterioration in Carbon Arc or Weatherometer, Recommended Practice for: (Atlas Weatherometer Model XW With Filters and Curved Racks).
D1052 - Measuring Rubber Deterioration, Method for (Cut Growth Using Ross Flexing Apparatus).
D1056 - Flexible Cellular Material Sponge or Expanded Rubber, Specification for.
D1564 - Flexible Cellular Materials Sponge or Expanded Rubber, Standard Method of Testing.
D1630 - Rubber Property Abrasion Resistance, Test for (NBS Abrader).
D1667 - Flexible Cellular Materials - Vinyl Chloride Polymers and Copolymers (Closed Cell Vinyl).
D2240 - Rubber Property Durometer Hardness, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard commercial product. The fenders shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturers's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the fenders being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish one complete fender of each classification included in the contract as a sample for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed may cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed herein to insure correct fit, assembly, and operations of the items. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Closed cell foam. The fender cushion shall be completely filled with a closed cell crosslinked polyethylene plastic foam of the characteristics listed in Table I.

3.4.2 Elastomer (urethane) shell. The elastomer shell encasing the fender cushion shall be poly-tetra-methylene glycol (PTMEG) polyether urethane elastomer, formulated to inhibit ultraviolet degradation, black in color and of the characteristics listed in Table II. In manufacturing the polyether urethane elastomer, 100 percent of the polyol used shall be PTMEG.

3.4.3 Steel. Steel for end fittings shall conform to ASTM A36.

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3.5 Design. The fender shall be designed as described herein.

3.5.1 General configuration. The fender shall have a cylindrical mid-body (that portion of the fender with a constant diameter) with conical or curved ends terminating in an end fitting located on the cylinders centerline. Size of fender shall be as described in 3.5.2.

TABLE I. Closed cell foam.

Density (ASTM D1667)	1.5 - 4.5 lb/cu ft
Tensile Strength (ASTM D1564 or D412)	35 lb/sq in (psi), minimum
Elongation (ultimate) (ASTM D1564 or D412)	150 percent, minimum percent to break
Continuous Service Temperature	-65 F to +175 F, minimum range
Water Absorption (after 24-hour immersion) (ASTM 1667)	0.07 lb/sq ft of cut surface, maximum
Tear Resistance (ASTM D624)	7 lb.in., minimum
25 Percent Compressive Resistance (ASTM D1667 or D1056)	6 - 22 psi minimum
50 Percent Compressive Resistance (ASTM D1667 or D1056)	14 - 35 psi minimum
25 Percent Compressive Set (ASTM D395 or D1667)	13 percent maximum

TABLE II. Urethane elastomer component of shell.

a.	Shore "A" Durometer Hardness (ASTM D2240)	85-92
b.	Tensile Strength (ASTM D412, Method A, Section 11-14)	2,500 psi minimum
c.	Elongation (ultimate) (ASTM D412, Method A, Section 11-14)	500 percent minimum
d.	Tear Strength (ASTM D470, Section 7.6)	100 lb/in.
e.	Flex Cut Growth (Ross, flexes to 5 X cut growth, ASTM 1052)	10,000 flexes minimum
f.	Abrasion Resistance NBS Abrader (ASTM D1630)	125 minimum
g.	Weatherometer Aging - 500 hours (ASTM D750)	Minimum 60 percent retention of original values of specification items (a) through (d) above

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3.5.2 Dimensions. The mid-body diameter, overall length, mid-body length, and skin thickness shall be as specified (see 6.2.1). Dimensions are given in Table III. The mid-body diameter, mid-body length, and overall length dimensions shall have a tolerance of ± 1 inch. The skin thickness values are given as a minimum. The fender length is measured from eye to eye of the end fittings.

TABLE III. Fender dimensions.

Diameter (ft)	Overall Length L(ft)	Minimum Mid-Body Length (ft)	Minimum Skin Thickness (in)	Minimum End Fitting Shackle Size (in)
3	6,8,10,12	L-2.1	7/8	3/4,3/4,7/8,7/8
4	8,10,12	L-3.0	1.0	1,1,1
5	10,14,16	L-3.5	1-1/4	1,1-1/4,1-1/4
6	12,14,16,18,20	L-4.6	1-1/4	1-1/4,1-1/4,1-3/8, 1-3/8,1-1/2
7	14,16	L-5.1	1-1/4	1-3/8,1-1/2
8	14,16,18	L-5.7	1-1/2	1-1/2,1-1/2,1-3/4
9	18,20	L-6.6	1-1/2	1-3/4,2
10	18,20	L-7.1	1-5/8	2,2

3.5.3 Repairability. The fender casing shall be repairable in the event of tears or punctures in the urethane shell. The repaired area shall have not less than 90 percent of the properties as specified in 3.4.2. Required repair materials shall be readily available from the fender manufacturer.

3.6 Construction. The fender shall be manufactured with a fabricated core of closed cell foam (see 3.4.1), encased in an elastomer shell (see 3.4.2). The fender shall have no exterior net or solid (non-compressible) interior longitudinal strength member connecting the fender end fittings.

3.6.1 Fender skin construction. The contractor shall certify that any seam of the fender is produced in such a manner as to assure a continuous homogeneous skin, with joint strength equal to that of the rest of the fender shell. Up to 15 filament reinforcing wraps of nylon tire cord may be used at the contractor's option. If filament wraps are used they shall be evenly distributed and the elastomer shall be applied in a continuous manner to assure adhesion between the various layers.

3.6.2 End fittings. The end fittings shall be connected by a chain and shall terminate in a clevis fitting sized for the shackle size as specified in 3.5.2 and Table III. The clevis fitting shall swivel to allow the end fitting to rotate freely on the axis of the fender. The end fittings shall be designed to transmit the ultimate load of the shackle to the fender. The end fittings shall be hot-dipped galvanized in accordance with ASTM A153 or ASTM A123.

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3.6.3 Repair kit. One repair kit shall be provided with the first fender delivered under a specific contract. The repair kit shall contain all materials and tools required for repairs of minor skin damage and instructions for it's use.

3.6.4 Operation and maintenance instructions. With the first fender delivered under a contract, the contractor shall furnish the Government instructions and precautions for handling, installing, and maintaining the fenders, normally furnished in commercial practice.

3.7 Performance. Fender performance, measured as energy absorption capacity and reaction force, shall be within ± 15 percent of the values listed in Table IV. Reaction force is defined as the load required to deflect the fender a given amount. The method of calculating required values for fender sizes not listed in the table is described in 6.5 and 6.6.

TABLE IV. Energy absorption capacity and reaction force of fenders.

Size Diameter X Length (ft)	E (minimum) (ft. kips) (See Note 1)	F (maximum) (kips) (See note 2)
3 X 6	39.0	48.0
3 X 8	55.0	68.0
3 X 10	71.0	88.0
3 X 12	87.0	108.0
4 X 8	89.0	83.0
4 X 10	118.0	110.0
4 X 12	146.0	136.0
5 X 10	180.0	134.0
5 X 14	269.0	200.0
5 X 16	313.0	233.0
6 X 12	300.0	186.0
6 X 14	364.0	226.0
6 X 16	427.0	265.0
6 X 18	491.0	305.0
6 X 20	554.0	344.0
7 X 14	487.0	259.0
7 X 16	574.0	305.0
8 X 14	619.0	288.0
8 X 16	733.0	341.0
8 X 18	847.0	394.0
9 X 18	1032.0	427.0
9 X 20	1175.0	486.0
10 X 18	1258.0	468.0
10 X 20	1435.0	534.0

1/ E = Energy absorption at 60 percent compression.
2/ F = Reaction force at 60 percent compression.

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3.8 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.9 Identification marking. The manufacturer's name, location, and fender serial number shall be lettered in white at both ends of the fender skin in a manner such that it can be read at a distance of ten feet. In addition, a metal tag or plate with stamped characters shall be attached to one end of the fender rigging providing the same information plus date of manufacture and the rated energy capacity of the fender.

3.10 Workmanship.

3.10.1 Urethane shell. The urethane shell of the fender shall be free from cracks, burrs, warpage, checks, chipped, or blistered surfaces and shall have a smooth surface.

3.10.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.10.3 Foam core. The foam core shall be homogeneous, fabricated from large sheets or slabs of foam, and bonded together to form a single constructed item consisting of as few pieces of foam as possible. All foam shall be as specified in Table I; the completed foam core density shall be within the limits specified in Table I. Chip, granular, or scrap foam shall not be used and the fabricated core shall be neither loose packed nor bound with a discontinous filler material.

3.10.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11 Inspection system. The contractor shall establish an inspection system in accordance with MIL-I-45208.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete fender when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall be performed on each complete fender in accordance with the examination of 4.3 and the packaging inspection of 4.5.

4.3 Examination. Each complete fender shall be examined for compliance with the requirements specified in section 3 of this specifiction. This element of inspection shall encompass all visual examinations and dimensional measurements. Records maintained in accordance with 3.11 shall be inspectd by the government to verify that the materials used in construction of all contract items conform to the requirements stated herein. In particular it shall be verified that each fender skin (shell) meets the material requirements of Table II and the minimum thickness of Table III, and that all the foam conforms to the requirements of sections 3.4.1 and 3.10.3. Average skin thickness may be estimated by the quantity of elastomer used for each fender. Any redesign or modification of the contractor's product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests. The first article shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.4.1 through 4.4.4 for fenders up to 7 feet in diameter. For fenders larger than 7 feet in diameter, due to the unavailability of suitable testing facilities, the tests of 4.4.1 and 4.4.2 shall not be performed. In lieu of such testing, the manufacturer shall verify the energy absorption capacity and reaction force requirements of Table IV by using results of prior tests on similarly constructed but smaller size fenders which it has built and tested (as outlined in 4.4.1) and extending them through engineering calculations to the larger sized fenders of this contract.

4.4.1 Energy absorption capacity and reaction force. The fender shall be compressed between two parallel flat plate surfaces to 40 percent of its original diameter (this shall be referred to as the 60 percent compression point) by an evenly distributed load, perpendicular to its longitudinal axis at a rate not less than 2.4 inches per minute. The load and corresponding deflection shall be recorded at 1-inch increments and plotted on a graph of load versus deflection. The load deflection curve shall then be integrated to generate an energy-deflection curve for the fender. At the 60 percent compression point on the energy-deflection curve, the energy absorption shall be not less than the value of E corresponding to that size fender in Table IV less the 15 percent tolerance. The reaction force, taken from the load-deflection curve at the 60 percent compression point or the compression point at which the full (without 15 percent tolerance reduction) E value of Table IV for that size fender is reached if occurring at less than 60 percent compression, shall not exceed the value of F for that size fender given in Table IV plus the 15 percent tolerance allowed. Upon release of the load, the fender shall show no sign of permanent distortion, rips, tears, delaminations, or other defects. Thirty minutes after release of the compression load, visible permanent deformation or elongation of the fender or compression set exceeding 4 percent (measured as reduction in fender diameter from that at the start of the test) shall constitute failure of the test. If the foam core is not bonded to the skin, measure the compression set by drilling 5 evenly spaced small holes, having an axis parallel to, the direction of load, along the crest of the top fender surface and measuring the change in distance to the foam from a suitable datum before compression and 30 minutes after release of the load.

4.4.1.1 Ultimate compression test. The procedure in 4.4.1 is continued and the fender is compressed until the 70 percent compression point is reached (without the plate surfaces contacting the end-fittings). The load shall then be released and the fender allowed to return to its original shape. There shall be no tears, delaminations, or permanent deformation which would reduce the mechanical integrity of the fender.

4.4.2 Strength test. The fender shall be compressed between two parallel flat plate surfaces and held at the working compression limit (60 percent) during the strength tests of 4.4.2.1 and 4.4.2.2. During these tests, the fender shall be restrained from sliding from between the compression plates by

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rigging a stop plate perpendicular to the direction of pull as shown in figure 1. Extrusion of the fender through the slot shall not constitute failure of the test. However, adequate safety precautions should be taken in the event extrusion of the fender occurs.

4.4.2.1 Longitudinal pull test. The stop plate and its opening shall be centered vertically and horizontally about the midpoint of the longitudinal centerline of the compressed fender. The opening in the stop plate shall have a height that is not less than 4 inches higher than the maximum diameter of the fender end fitting or elastomer collar. The opening shall have a minimum width that is 2 inches wider than the width of the compressed fender. The end fitting opposite to the stop plate shall be free during the test. The end fitting adjacent to the stop plate shall be subject to a tensile load applied in line with the longitudinal axis of the compressed fender. For fenders greater than 4 feet in diameter, the load shall be applied in 10,000 pound (lb) increments until a load equal to 90,000 lb is achieved. For fenders 4 feet in diameter or less, the load shall be applied in 5,000 lb increments until a load equal to 50,000 lb is achieved. Each incremental load shall be held for 1 minute and the final load held for 5 minutes. The loads shall then be released and the fender allowed to return to its original shape. There shall be no tears, delaminations, or permanent deformation which would affect the mechanical integrity of the fender. The end fitting shall show no sign of permanent deformation or separation from the fender.

4.4.2.2 Transverse pull test. A stop plate shall be centered between the two ends of the compressed fender and the plate shall be such that it will not interfere with the end fittings or the elastomer collars during the test. Both end fittings shall be simultaneously subjected to a tensile load applied perpendicular to the longitudinal axis of the fender and in a horizontal plane parallel to the plates shown in figure 1. For fenders greater than 4 feet in diameter, the load shall be applied simultaneously to each end fitting in 10,000 lb increments (except for the last increment) up to a load on each end fitting equal to 45,000 lb. For fenders 4 feet in diameter or less, the load shall be applied simultaneously to each end fitting in 5,000 lb increments up to a load on each end fitting equal to 25,000 lb. Each incremental load shall be held for 1 minute and the final load held for 5 minutes. The loads shall then be released and the fender allowed to return to its original shape. There shall be no tears, delaminations, or permanent deformation which would affect the mechanical integrity of the fender. The end fittings shall show no sign of permanent deformation or separation from the fender.

4.4.3 Fender sustained load test. This test shall be performed as an option (see 6.2.1). Compress the fender as described in 4.4.1 to a load equal to that required to obtain the rated energy obtained in 4.4.1 and hold this load for one hour. Record load and deflection at the beginning and end of this one hour period and at least once in between at about the 30 minute point. Failure of the fender or its foam core to rebound to 90 percent of its diameter one hour after the release of the load shall constitute failure of this test. Follow the procedures of 4.4.1 for measuring rebound of permanent set.

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4.4.4 Urethane elastomer retains. Test coupons of the urethane elastomer used to manufacture each fender shall be identified and retained by the contractor for a period of not less than 5 years from the date of fender manufacture. A coupon from every third fender of the lot shall be submitted to an independent laboratory for physical testing in accordance with 3.4.2 and Table II, (a) through (d). An example of coupon selection is fender 1, 4, 7, 10, 13, etc. The certified test results of these tests are to be submitted to the Cognizant Contract Quality Control Representative for approval. The government reserves the right to request any of the remaining specimens within the 5 year period and perform any or all of the tests of 3.3.2 at government expense. The test coupons shall be 36 square inches (sq. in.) minimum by .09 ± .03 in. thick.

4.4.4.1 Collection fixture.

a. If the fender skin is cast, the fixture shall be compatible with the casting technique used and of the above miniumum dimensions.
b. If the fender skin is built up by spraying, the fixture shall be a flat plate of 36 sq. in. area or greater.

4.4.4.2 Fixture preparation. The appropriate fixture with release agent applied is to be at the same temperature as the fender skin when formed.

4.4.4.3 Coupon collection.

a. If the fender skin is cast, the coupon shall be formed from liquid urethane in the same manner and timing that the fender is cast.
b. If the fender skin is built-up by spraying, the collection fixture is to be passed through the spray stream so that one layer, equivalent to one layer of urethane on the fender shell, is applied. This procedure is to be repeated until the specified coupon thickness is achieved. The time between coats is to be the same as the time between layers applied to the fender shell.

4.4.4.4 Coupon curing. The coupon is to be cured under the same conditions as the fender shell.

4.4.4.5 Frequency of collection. One or more coupons are required for each fender shell.

a. If cast in a mold, one coupon is required for each component cast, i.e., ends, cylinder, joints, etc.
b. If sprayed, a minimum of one coupon, representative of the skin cross section, is required.

4.5 Packaging inspection. The packing and marking of the fenders shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

5.1 Packing. Packing shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. Fenders shall be packed in crates conforming to MIL-C-3774, type I, style A. Anchoring, cushioning, blocking, bracing, and waterproofing with a full shroud shall be in accordance with the appendix of MIL-C-3774.

5.1.2 Level C. Fenders shall be packed in accordance with the contractor's standard practice in a manner to prevent corrosion, deterioration, and damage and to insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Fenders described in this specification are used to protect the hull of vessels when moored side by side or against piers and wharves.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When the issue of documents referenced herein shall be other than those listed in the issue of DODISS specified in the solicitation (see 2.1.1).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When the optional test is required (see 4.4.3).
e. Dimension of fender required (see 3.5.2).
f. Level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Energy absorption capacity. The energy absorption capacity for various size fenders of the same diameter is directly proportional to fender length assuming the fenders are cylinders having diameter and length equal to the respective nominal dimensions of the actual fender.

6.6 Reaction force. The reaction force for different size fenders of the same diameter is directly proportional to the fender mid-body length.

6.7 Key word listing.

Foam filled fenders
Fenders
Fendering system
Camels
Berthing facilities

Preparing Activity:

Navy - YD

(Project 2040-0156)

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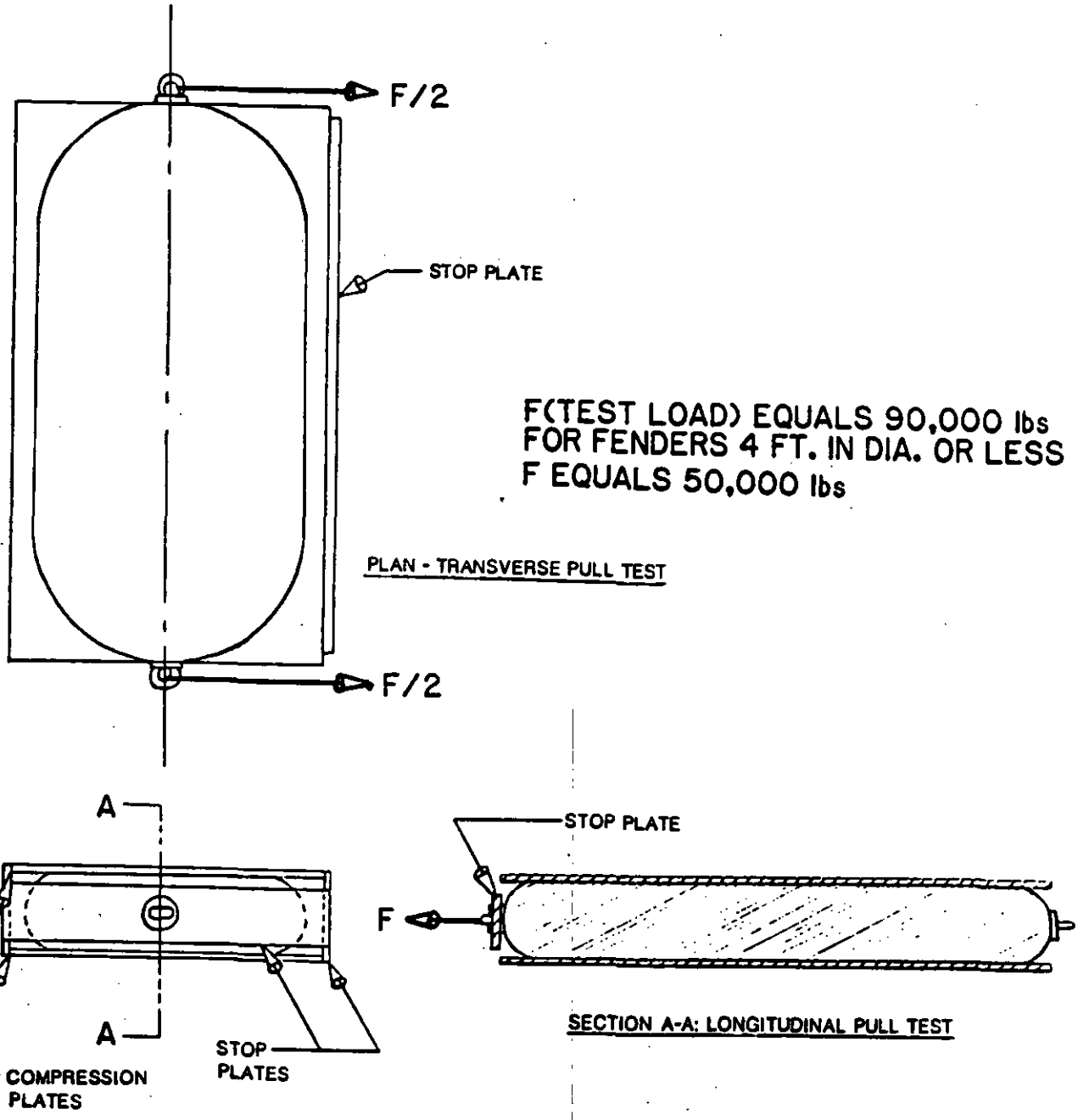


FIGURE 1. PULL TEST ARRANGEMENT (SCHEMATIC)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-29248 (YD)	FENDERS, MARINE, FOAM FILLED, NETLESS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION (YYMMDD)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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